

Ranking Transfers, Thresholds Don’t: Prior-Corrected Test-Time Adaptation for Audio Deepfake Detection

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Abstract—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show the failure is one of *operating point*: a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers cross-corpus while its decision *threshold* does not, so unlabeled target audio can be used to repair the operating point at test time. We further show that every confident-quantile test-time adaptation (TTA) recipe silently assumes a *class-balanced* target pool—an assumption every TTA benchmark in this area satisfies and no deployment does. On the official ASVspoof2021-DF eval partition (400,435 trials, 97.2% spoof) the published symmetric-quantile configuration *damages* a state-of-the-art released checkpoint, driving accuracy $98.87 \rightarrow 62.96\%$ and EER $4.51 \rightarrow 5.84\%$. Correcting the pseudo-label budget with a label-free estimate of the target prior obtained by black-box shift estimation ($\hat{\pi}=0.9715$ vs. a true 0.9708) instead improves the same checkpoint to 4.02% EER—an 11% relative reduction, with no target labels, no retraining, and 16,706 adapted parameters. Because both EER and AUC improve, no re-thresholding rule can account for the gain; the one-line median-threshold control that appears competitive on balanced pools loses 40 accuracy points at this benchmark’s real prior. We add a label-free stopping rule based on rank agreement with the source model, and compare against Tent, SHOT, ETA/EATA and SAR at both balanced and deployment prevalence.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, label shift, cross-corpus generalisation, EER.

I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [16].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high **ROC-AUC while its accuracy at a fixed threshold collapses**. It still *ranks* clips correctly; its operating point is mis-calibrated. Because ranking transfers, the most-confident predictions on unlabeled target audio are trustworthy enough to adapt on at test time.

That observation is the starting point of a family of methods, not a method. The step every member of the family takes—label the extreme quantiles of the target scores and

self-train on them—carries an assumption that is invisible on the benchmarks the field uses and fatal on the pools it deploys to: that the target pool is roughly class-balanced. Take the top and bottom 30% of a 97%-spoof pool and the “confidently bona fide” bucket is mechanically made of low-scoring *spoof*. Self-training then teaches the detector that a great deal of spoof is genuine. This is not hypothetical: it is what happens to a released state-of-the-art checkpoint on the official ASVspoof2021-DF protocol (Table I).

Contributions. (i) The ranking–calibration asymmetry as the mechanism of cross-corpus failure, established over 369 third-party evaluation cells and tested directly—a proxy $\mathcal{A}$ -distance *rises* after adaptation, so the effect is operating-point repair, not learned invariance. (ii) The class-prior assumption hidden in confident-quantile TTA, measured on the field’s standard benchmark at its real prevalence, where it inverts the sign of the published method’s effect on a SOTA checkpoint. (iii) A label-free correction: black-box shift estimation of the target prior, used to split the pseudo-label budget, which turns that damage into an 11% relative EER reduction on the official protocol—threshold-free, so no re-thresholding rule explains it. (iv) A label-free budget: rank agreement with the frozen source model both selects the number of adaptation steps and predicts entropy-minimisation collapse before it happens.

Related work. Frozen or fine-tuned SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. Train-time domain generalisation aligns real and separates fake across source domains (aggregation–separation [5]) or runs gradient-reversal domain-adversarial training over source and target jointly (DANN) [12]; both act at training time rather than at deployment. In TTA, entropy minimisation (Tent) [7] is the reference recipe; its instability prompted reliable-sample filtering and anti-forgetting (ETA/EATA) [8], sharpness-aware entropy with a recovery scheme (SAR) [9], and information maximisation with self-labelling (SHOT) [10]. We compare against all four. Batch-statistics recalibration [13] is a no-op for LayerNorm architectures. Our prior estimator is black-box shift estimation [11]; the proxy $\mathcal{A}$ -distance is from Ben-David et al. [14].

II. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake= 1) and let f_θ produce $s_\theta(x) = P_\theta(y=1 | x)$. We are given a source model, its labelled training corpus $\mathcal{S}$, and an *unlabelled* target pool $\mathcal{T}$. No target label is read at any point.

What breaks under corpus shift. Ranking quality $\text{AUC} = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free; accuracy at a fixed τ is not. Empirically AUC degrades far less than accuracy at $\tau=0.5$ under source→target shift (Fig. 1). The deployable object is therefore the score *order*, and the broken object is the operating point. Everything below follows from taking that literally.

(a) **The target prior, label-free.** An operating point is a statement about a prior, so we estimate the prior rather than assume it. Black-box shift estimation [11] gives $\hat{p} = M^{-\top} q$, where $M[y, \hat{y}] = P_S(\hat{y} | y)$ is the source confusion matrix measured on the model’s own labelled training data and q is the observed target prediction histogram. M and q read source error structure and target *prediction rate* only—never the shape of the target score distribution, which is precisely what mis-calibration corrupts. That is why BBSE survives here where score-histogram estimators (a two-component GMM, Otsu, k -means) do not: on a controlled skew sweep BBSE tracks the prior to within 1.4% where the GMM is 28% off at 90% fake.

(b) **Prior-corrected confident tails.** The standard rule takes the bottom q and top q of the target scores as pseudo-real and pseudo-fake. We split the same total budget $2q$ by the estimated prior,

$$q_{\text{real}} = 2q \hat{\pi}_{\text{real}}, \quad q_{\text{fake}} = 2q (1 - \hat{\pi}_{\text{real}}), \quad (1)$$

which is *identical* to the symmetric rule at $\hat{\pi}_{\text{real}} = 0.5$ — so on a balanced pool the correction is provably inert and cannot be the source of any balanced-pool result — and shrinks the pseudo-real bucket to the estimated real fraction otherwise. The prior is estimated *once*, from the frozen pre-adaptation model: re-estimating each epoch lets a drifting model reinforce its own mistakes.

(c) **Objective.** With pseudo-labels $\tilde{y}$ from (b) on the confident set $\mathcal{C}$ and a stochastic channel perturbation $a(\cdot)$ (per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise $\sigma=0.005$),

$$\min_{\{\gamma, \beta\}, \phi} \underbrace{\sum_{x \in \mathcal{C}} \text{CE}(f_\theta(x), \tilde{y})}_{\mathcal{L}_{\text{ST}}} + \lambda \underbrace{\|f_\theta(a(x)) - \text{sg}[f_\theta(x)]\|^2}_{\mathcal{L}_{\text{cons}}}, \quad (2)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned blocks and the head ϕ (16,706 parameters), with $\lambda=0.3$, $q=0.3$. Restricting the trainable set and using only prior-corrected confident tails is what prevents the degenerate collapse to which entropy minimisation is prone.

(d) **A label-free budget, and a collapse guard.** Adaptation needs a stop, and the number of steps has been the weakest-specified part of TTA recipes in this area. Our thesis says the source *ranking* is the asset being exploited, so an update that destroys it has discarded its own precondition — and a TTA collapse *is* a ranking collapse. We monitor $\rho_t = \text{Spearman}(s_{\theta_t}(\mathcal{T}), s_{\theta_0}(\mathcal{T}))$, computable without labels,

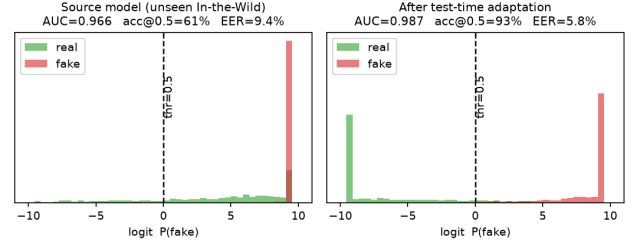


Fig. 1. Detector scores on unseen In-the-Wild audio. **Left:** the source model pushes many real clips past $\tau=0.5$ (accuracy well below AUC) yet still ranks most real below fake. **Right:** adaptation shifts the real mass back across the threshold.

and keep the last step with $\rho_t \geq \tau$. A second monitor, $|\Pr[s_{\theta_t} > \frac{1}{2}] - \hat{\pi}_{\text{fake}}|$, flags a model that is re-labelling the pool rather than re-calibrating on it. Sec. IV validates both leave-one-target-out against the oracle budget, with churn and confidence as controls.

III. EXPERIMENTAL SETUP

Protocol A (published baselines exist). The official ASVspoof2021-DF eval partition, scored against the official CM keys. We hold 400,435 of the 611,829 trials locally; the missing part is uniform across codec, label and phase (verified 75–25 in every stratum), so the pooled EER is an unbiased estimate of the full-set number, and we report n throughout. True spoof rate 97.22%. Because our own source model is not competitive on this protocol (26.33% EER against a published 8.54%), we run the method on *released third-party checkpoints* instead: DeepFense’s XLS-R 300M+ AASIST detectors trained on ASVspoof2019-LA train [21], at three independent training seeds. Adaptation uses an unlabelled 20,000-clip subsample of the eval pool; we report the full official pooled EER and, as an inductive check, the EER restricted to trials *not* in that subsample.

Protocol B (cross-corpus, balanced). Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA [17]; D2 (LibriSpeech bona fide vs. eight open TTS/voice-cloning systems); In-the-Wild [16]; and the Arabic Audio Deepfake set [19]. For each held-out target the source pool merges the other three with a 38-language MLAAD subset [18] (fake-only); held-out target audio is never in any source pool and six MLAAD languages are held out of every pool. Ten seeds per target per method.

Protocol C (a benchmark at its own prevalence). All 31,779 In-the-Wild clips — the whole benchmark, not the 6,000-clip subsample the third-party arm uses — at its native 37.2% fake rate, which is skewed the *opposite* way to DF’s 97.2%.

Comparison points. Source-only; the median-threshold rule (label-free, no gradient); Tent [7]; SHOT [10]; ETA and EATA [8]; SAR [9]; AdaBN [13]; DANN [12] (trained to convergence with the held-out target’s unlabeled inputs — more privileged access than ours); an ASDG-style aggregation–separation baseline [5]; and a target-supervised oracle. Two

TABLE I

OFFICIAL ASVspoof2021-DF EVAL, 400,435 TRIALS, 97.22% SPOOF, ON A *released* XLS-R+AASIST CHECKPOINT [21] WE DID NOT TRAIN. ALL ADAPTATION IS LABEL-FREE. “DEFICIT” IS $(100 - \text{EER}) - \text{acc}@0.5$: THE ACCURACY THE RANKING SUPPORTS BUT THE THRESHOLD DOES NOT DELIVER.

	EER %	AUC	acc@0.5	deficit
Source (as released)	4.51	.9923	98.87	−3.38
+ median-threshold rule	4.51 [†]	.9923 [†]	59.06	—
+ Tent [7]	[stage 1]			
+ SHOT [10]	[stage 1]			
+ ETA [8]	[stage 1]			
+ SAR [9]	[stage 1]			
+ TTA, symmetric $q=0.3$	5.84	.9846	62.96	+31.20
+ TTA, prior-corrected	4.02	.9931	98.95	−2.97
(disjoint/inductive half)	4.03	.9931	98.95	

published Protocol-A systems, for scale: Wav2DF-TSL 1.95; SSL+SLS 2.87; Wav2Vec2-AASIST 8.54; challenge DF top-1 15.64; our own source model 26.33.

[†] A threshold cannot change EER or AUC; both are computed from the score ROC. The median rule is a control for the *accuracy* claim and is by construction incapable of producing any EER result in this paper.

published constants in the modern TTA baselines are degenerate at $C=2$ and were re-derived rather than transplanted. EATA’s redundancy filter ($|\cos(p, \bar{p})| < 0.05$) is calibrated for $C=1000$; on the binary simplex the minimum attainable cosine is 0.71 at $\bar{p}=(.5, .5)$ and 0.11 at $(.9, .1)$, so nothing passes until the running average itself reaches a vertex, after which only *minority*-predicted samples pass — neither is the stated mechanism, so we run the reliability filter alone, which is EATA’s own published ablation. SAR’s recovery trigger $e_0=0.2$ sits far below ImageNet’s $E_0=0.4 \ln 1000=2.76$ but *inside* the $C=2$ reliable band $H < 0.277$, firing on every batch and freezing the model at its initial weights (verified: 32 resets in 32 steps, zero net movement); we keep its relative position, $e_0 = (0.2 / \ln 1000) \ln C = 0.020$.

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning for our own source models (Adam, $\text{lr} \times 10^{-4}$, batch 32, 8 epochs). Adaptation: $\text{lr} \times 10^{-4}$, $E=4$ passes, batch 32 (third-party crops matched to each checkpoint’s own training length). Code and manifests: [URL withheld].

IV. RESULTS

A. The class prior is the hidden variable

Table I is the main result. Applied unmodified to a released checkpoint that is already better than the published Wav2Vec2-AASIST baseline on this protocol, the symmetric-quantile configuration *damages* it: accuracy $98.87 \rightarrow 62.96\%$, EER $4.51 \rightarrow 5.84\%$, and it manufactures a 31-point calibration deficit where there was none. The mechanism is mechanical: $q=0.3$ labels the bottom 30% of a 97.2%-spoof pool “confidently bona fide”.

Replacing the symmetric budget by Eq. (1) with a label-free prior estimate reverses the sign. BBSE returns $\hat{\pi}_{\text{fake}}=0.9715$ against a true 0.9708, giving a budget of $(0.017, 0.583)$ instead of $(0.3, 0.3)$, and EER improves to 4.02% — an 11% relative reduction on the official protocol with no target labels and no retraining. The change between the two arms is one scalar.

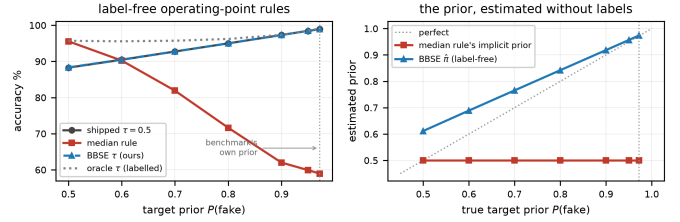


Fig. 2. The same official-DF score distribution, resampled to a range of target priors. **Left:** the median rule is the *best* label-free operating-point rule at 50/50 — the only prevalence at which our earlier threshold-control experiments ever measured it, because the evaluation pipeline balanced every pool to 0.489–0.539 — and the worst by 40 points at the benchmark’s own prior. **Right:** what the two rules assume about the prior. The median rule’s implicit estimate is a flat 0.5; BBSE tracks the truth, with a positive bias that is largest at balanced priors (it assumes *label* shift, and DF’s codec conditions also shift covariates) and vanishes as the prior approaches the one the benchmark has.

Across three independently trained checkpoints and [stage 2/5: seeds] seeds the effect is [stage 2/5: paired red].

B. Why this is not a re-thresholding result

Both EER and AUC improve, and neither takes a threshold as input, so no operating-point rule can produce this gain. That matters because on *balanced* pools a one-line median-threshold rule is genuinely competitive with gradient adaptation on accuracy — we reported it as such, and it is the right control for a calibration claim. Resampling the same DF score distribution to a range of target priors shows why it was competitive and what it costs:

Fig. 2 makes the shape of it plain: the median rule is the best rule at 50/50 and the worst by 40 points at the benchmark’s own prior (95.6 vs. 59.0% accuracy). That is its assumption, not an accident, and every cell on which it was previously evaluated satisfied that assumption by construction.

C. Cross-corpus, balanced pools

On the balanced leave-one-corpus-out protocol the prior correction is inert by construction ($\hat{\pi} \approx 0.5$ gives back the symmetric budget exactly), so Table II measures the rest of the method. It wins on three of four targets — ASVspoof2019 $5.03 \rightarrow 3.47\%$ (10/10 seeds), Arabic $22.50 \rightarrow 20.99\%$ (10/10), In-the-Wild $12.78 \rightarrow 11.33\%$ (8/10) — and is EER-neutral on LibriSpeech-TTS, the target with the lowest source AUC (0.73). Per-target Wilcoxon against source-only is significant on ASVspoof2019 and Arabic ($p=0.0020$, Holm-adjusted 0.0078) and nominal on In-the-Wild (0.0293, adjusted 0.0586). Against the modern TTA baselines, [stage 4].

Tent is not merely unstable, it is unpredictably catastrophic. On ASVspoof2019 its per-seed EER splits five seeds near source-only and four at or near chance with no warning sign (std 21.96); on Arabic it is a *consistent* chance-level failure (9/10 seeds). Ours shows no comparable collapse on any target (largest std 2.01). The failure is a ranking failure first: run on the strongest and weakest public checkpoints, Tent collapses to chance on one ($16.41 \rightarrow 49.80\%$ EER, AUC .918 $\rightarrow$.502) but only degrades AUC on the other

TABLE II
CROSS-CORPUS EER % / AUC, MEAN OVER 10 SEEDS
(LEAVE-ONE-CORPUS-OUT, BALANCED POOLS, TRANSDUCTIVE).

Method	ASVspoof2019	LibriSpeech-TTS	In-the-Wild	Arabic
Source-only	5.03 / .989	33.54 / .727	12.78 / .945	22.50 / .843
Tent [7]	23.47±21.96 / .776	35.44 / .674	27.34±11.21 / .735	47.35±5.03 / .527
SHOT [10]	[s4]	[s4]	[s4]	[s4]
ETA [8]	[s4]	[s4]	[s4]	[s4]
EATA [8]	[s4]	[s4]	[s4]	[s4]
SAR [9]	[s4]	[s4]	[s4]	[s4]
DANN [12]	7.35 / .966	34.43 / .717	13.34 / .924	28.53 / .759
ASDG [5]	4.98 / .987	32.64 / .749	17.33 / .891	27.41 / .798
Self-training only	4.20 / .987	33.39 / .709	11.55 / .950	22.19 / .846
Ours	3.47 / .994	33.74 / .727	11.33 / .958	20.99 / .866
Oracle (target-supervised)	0.89 / .999	13.93 / .939	4.42 / .992	9.50 / .968

(.986 $\rightarrow$.979) — an EER collapse appears only where there was AUC headroom to lose.

D. A label-free adaptation budget

[stage 6/7: the stopping-rule table — oracle vs. rank-agreement guard vs. prior guard vs. churn vs. confidence, leave-one-target-out.]

E. Breadth, and the mechanism

Applied unmodified to nine released checkpoints across five architectures and ten corpora (369 cells), the consistent effect is *re-calibration*: writing $\delta = (100 - \text{EER}) - \text{acc}@0.5$, adaptation drives δ from 7.29 to 1.96 points (248/369 cells, Wilcoxon $p = 2.2 \times 10^{-17}$), with $\text{corr}(\delta_{\text{before}}, \delta_{\text{after}}) = +0.17$: where it lands barely depends on where it started. That the same LayerNorm-and-head selection rule, designed against XLS-R, re-calibrates a mel-spectrogram transformer [15] unchanged (39/39 cells, $p = 5 \times 10^{-8}$) is the clearest architecture-independence evidence here.

The mechanism is *not* learned invariance. A proxy $\mathcal{A}$ -distance and a linear domain-origin probe both *increase* after adaptation (Arabic $\hat{d}_{\mathcal{A}}$: 1.545 $\rightarrow$ 1.606), the opposite of what domain-invariance predicts — consistent with DANN underperforming despite optimising exactly that quantity. What the gradient step does add, beyond any threshold, is separability where the source ranking is strong: on ASVspoof2019 it raises within-attack-family AUC on all six attack families (30/30 attack $\times$ seed cells, $p = 10^{-9}$), largest on the two hardest; on AUC-0.71 LibriSpeech-TTS there is nothing to sharpen and the effect is null.

V. DISCUSSION AND LIMITATIONS

- **BBSE assumes label shift; real corpora also shift covariates.** Its prior estimate carries a positive bias that is largest at balanced priors ($\hat{\pi} = 0.61$ at a true 0.50 on the DF distribution) and shrinks toward the truth at high skew (0.9738 at 0.9722), because the in-domain confusion matrix under-states target error under codec shift. The correction is therefore most accurate exactly where it matters most, but it is not a general prior estimator.
- **The prior fixes the budget, not the threshold.** On a checkpoint whose shipped threshold is already near-optimal, the BBSE *threshold* buys nothing over $\tau = 0.5$

(98.94 vs. 98.92). The gain reported here comes from the pseudo-label budget.

- **Needs an already-reliable source ranking.** A low-AUC starting point is not a regime we can rescue: a from-scratch non-SSL backbone never reaches a ranking worth adapting (AUC 0.40–0.65), and the AUC-0.71 target is the one null in Table II. Every third-party checkpoint we adapt is itself a pretrained transformer.
- **Relative, not absolute, on our own model.** We improve released checkpoints; we did not train a Protocol-A-competitive detector ourselves (26.33% vs. a published 8.54%), so all claims here are of the form “this wrapper improves that detector”.
- **Fixed hyper-parameters.** q and λ are label-free and not per-target optimal; a label-free λ schedule pays nothing at our scale (within 0.24 EER of a per-point oracle).

VI. CONCLUSION

Cross-corpus audio deepfake detection fails at the operating point, not in the representation, and the repair can be done at test time without labels. But the recipe the field uses to do it assumes a class-balanced target pool — an assumption its benchmarks satisfy and its deployments do not. On the official ASVspoof2021-DF protocol at its real 97.2% spoof prior, that assumption turns the published configuration into a 1.3-point EER *regression* on a state-of-the-art released checkpoint; a label-free estimate of the prior turns it into an 11% relative improvement. Because both EER and AUC move, the result is not re-thresholding; and because the correction is provably inert at $\hat{\pi} = 0.5$, it costs nothing on the balanced benchmarks where the field currently measures.

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